AUTONOMOUS AI MARKETING AGENT

Complete Technical Blueprint

How to Build an AI System That Finds Customers
and Delivers Products to Them Autonomously

The name is (OUTRIQ)

Without Google Ads. Without Meta Ads. Without Manual Campaigns.

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1. Product Overview & Vision

1.1 What This Product Is

This document describes the complete technical architecture for an Autonomous AI Marketing Agent — a system where businesses input their product information once, and artificial intelligence handles the entire process of finding potential customers and placing the product in front of them across the internet.

The system eliminates the need for businesses to understand ad platforms, manage campaigns, set budgets, choose keywords, or monitor performance dashboards. The AI acts as an autonomous marketing department that works around the clock.

1.2 How It Differs from Google Ads and Meta Ads

Dimension	Google/Meta Ads	AI Marketing Agent
Setup Complexity	High — requires campaign structure, ad groups, targeting, budgets, creative assets	Minimal — product name, description, image, target market
Cost Model	Pay per click or impression. Costs escalate with competition.	Pay for compute and API usage. Fraction of traditional ad spend.
Distribution	Limited to owned platforms (Google Search, YouTube, Instagram, Facebook)	Goes wherever the audience gathers: forums, social media, blogs, search, AI platforms
Management	Requires ongoing human optimization	Fully autonomous after initial setup
Intelligence	Keyword and demographic matching	Deep semantic understanding of product-to-customer fit
Reach	Platform-locked	Platform-agnostic — reaches across the open internet

1.3 Core Product Promise

A business owner enters their product details. The AI takes over. Within hours, the product begins appearing in relevant conversations, search results, content recommendations, and digital spaces where their ideal customer already spends time. No ad budget required. No marketing expertise needed.

2. System Architecture Overview

The system operates on three core layers, each handling a distinct phase of the autonomous marketing cycle. Every layer communicates through a central orchestration engine that coordinates timing, prioritization, and feedback loops.

2.1 The Three Layers

Layer 1: Business Intake System

This is the entry point. Businesses register, enter their product information, and the AI processes it into a deep understanding of what the product is, who needs it, and why. This layer transforms raw business input into a structured intelligence profile that the rest of the system uses.

Layer 2: Intelligence & Matching Engine

The brain of the system. This layer continuously scans the internet for signals of customer intent — questions being asked, problems being described, needs being expressed — and matches those signals to registered products. It decides where to act, what to say, and when to engage.

Layer 3: Autonomous Distribution Engine

The execution arm. This layer takes the decisions from the Intelligence Engine and carries them out — generating content, posting responses, building web pages, sending outreach, and placing the product in front of the right people in the right context at the right time.

2.2 System Data Flow

1. Business enters product information through web dashboard or API.
2. Intake processor enriches and structures the data into a Product Intelligence Profile.
3. Intelligence Engine runs continuous scans across monitored channels for intent signals.
4. Matching algorithm pairs intent signals with relevant Product Intelligence Profiles.
5. Distribution Engine selects the optimal channel and action type for each match.
6. AI generates the content, message, or page appropriate for that channel.

7. Action is executed (post published, response sent, page deployed, outreach delivered).
8. Feedback loop captures results (clicks, responses, conversions) and feeds them back to the Intelligence Engine to improve future matching.

2.3 Technology Stack Summary

Component	Technology	Purpose
Backend Framework	Python (FastAPI) or Node.js (Express)	API server, business logic, orchestration
AI/LLM Layer	Claude API (Anthropic)	Content generation, intent analysis, product matching
Database	PostgreSQL + Redis	Product profiles, user data, caching, job queues
Vector Database	Pinecone, Weaviate, or ChromaDB	Semantic search and product-to-intent matching
Task Queue	Celery (Python) or BullMQ (Node.js)	Scheduling and managing autonomous distribution tasks
Web Scraping	Playwright, Puppeteer, or Scrapy	Social listening, forum monitoring, search analysis
Content Hosting	Vercel, Netlify, or AWS S3 + CloudFront	Hosting auto-generated landing pages and content
Monitoring	Grafana + Prometheus	System health, task success rates, compliance tracking
Frontend	React or Next.js	Business dashboard for product management and analytics

3. Layer 1: Business Intake System

3.1 What the Business Provides

The intake form should be as simple as possible. Every additional field is friction that reduces signup completion. The AI should infer as much as possible from minimal input.

Required Fields

Field	Type	Example	Why It Matters
Business Name	Text	ChopFit Lagos	Identity for all generated content
Product/Service Name	Text	Healthy Meal Prep Delivery	Core offering the AI will market
Description	Long Text (500 words max)	We prepare fresh, calorie-counted meals delivered to your door in Lagos...	The AI's primary source for understanding the product
Target Customer	Text	Health-conscious professionals aged 25-40 in Lagos	Defines who the AI should look for
Location/Service Area	Text or Dropdown	Lagos, Nigeria	Geographic targeting for distribution
Website URL	URL (optional)	www.chopfit.ng	Destination for generated traffic
Product Image	Image upload (optional)	Photo of meal prep boxes	Used in generated content and landing pages
Category	Dropdown	Food & Beverage / Health & Fitness	Helps the matching engine narrow its search

3.2 Product Intelligence Profile Generation

Once the business submits their information, the AI immediately processes it into a Product Intelligence Profile (PIP). This is the internal document the entire system references. The PIP is not visible to the business — it powers everything behind the scenes.

How the PIP is Generated

1. The product description is sent to the LLM (Claude API) with a structured prompt that extracts: core value proposition, primary pain points the product solves, ideal customer psychographic profile, key differentiators from competitors, emotional triggers that would make someone want this product, relevant search terms and phrases a potential customer might use.
2. If a website URL was provided, the system crawls the website using Playwright or Puppeteer, extracts all text content, and feeds it to the LLM as additional context.
3. The LLM output is structured as JSON and stored in the database alongside the original business input.
4. A vector embedding of the PIP is generated using an embedding model (such as OpenAI's text-embedding-3-small or Cohere's embed-english-v3.0) and stored in the vector database for semantic matching.

Example PIP Structure (JSON)

```
{
  "business_id": "chopfit-lagos-001",
  "product_name": "Healthy Meal Prep Delivery",
  "value_proposition": "Saves busy professionals time while keeping them
healthy",
  "pain_points_solved": [
    "No time to cook healthy meals",
    "Struggling to maintain diet with busy schedule",
    "Tired of unhealthy fast food options"
  ],
  "ideal_customer": {
    "age_range": "25-40",
    "location": "Lagos, Nigeria",
    "psychographic": "Career-driven, health-conscious, values convenience",
    "platforms": ["Instagram", "X/Twitter", "Nairaland", "LinkedIn"]
  },
  "search_terms": [
    "healthy food delivery Lagos",
    "meal prep service Nigeria",
    "diet food delivery Lekki",
    "calorie counted meals Lagos"
  ],
  "emotional_triggers": [
    "Feeling guilty about eating junk food again",
    "Wanting to look and feel better",
    "Frustration with wasted time cooking"
  ],
```

```
  "content_angles": [
    "What if your meals were handled for the whole week?",
    "Lagos professionals are switching to this and seeing results",
    "Stop spending 2 hours cooking what takes us 5 minutes to deliver"
  ]
}
```

3.3 Technical Implementation

API Endpoint Design

```
POST /api/v1/products
  Body: { name, description, target_customer, location, category, website_url,
image }
  Response: { product_id, status: 'processing', estimated_ready: '2 minutes' }
```

```
GET /api/v1/products/{product_id}/profile
  Response: { pip: { ...full Product Intelligence Profile }, status: 'active' }
```

```
GET /api/v1/products/{product_id}/performance
  Response: { impressions, engagements, clicks, leads, channels_active,
top_performing_channel }
```

Background Processing Pipeline

When a product is submitted, the system queues the following tasks in order:

1. Validate and sanitize input data.
2. If website URL provided: spawn a headless browser to crawl and extract content (max 10 pages).
3. Send combined product data + website content to LLM with PIP generation prompt.
4. Parse LLM JSON response and store in PostgreSQL.
5. Generate vector embedding and upsert to vector database.
6. Mark product as 'active' — the Intelligence Engine can now match it.
7. Trigger initial content generation tasks (landing page, first batch of SEO articles).

4. Layer 2: Intelligence & Matching Engine

This is the core brain of the system. It continuously monitors the internet for signals that someone needs a product, then matches those signals to registered products in real time.

4.1 Intent Signal Detection

An intent signal is any online activity that suggests someone is looking for, needs, or would benefit from a product. The Intelligence Engine monitors multiple signal sources.

Signal Source 1: Social Media Monitoring

The system monitors public posts, comments, and discussions on platforms where the target audience gathers. It watches for posts that express a need, ask a question, describe a problem, or request a recommendation that matches a registered product.

How it works: A scheduled task runs at configurable intervals (default: every 15 minutes per channel). It uses platform APIs where available (X/Twitter API, Reddit API) and web scraping where APIs are insufficient. Each post is analyzed by the LLM to determine: (a) does this express a need that matches any registered product? (b) what is the confidence level of the match? (c) what is the best way to engage — direct response, content share, or passive reference?

Platforms to Monitor

Platform	Access Method	What to Monitor	Signal Examples
X/Twitter	X API v2 (search endpoint)	Posts, replies, hashtags, trending topics	"Anyone know a good meal prep service in Lagos?"
Reddit	Reddit API (PRAW library)	Subreddit posts and comments	"Looking for healthy food delivery in Nigeria"
Nairaland	Web scraping (Playwright)	Forum posts and discussions	"Where can I find calorie counted food in Lekki?"
Facebook Groups	Web scraping (careful rate limits)	Group posts and comment threads	"Please recommend a healthy meal service in Lagos"
Quora	Web scraping or Quora API	Questions and answers	"What is the best meal delivery service in Lagos?"
Instagram	Instagram Graph API (limited)	Comments, hashtag searches	Comments under fitness posts asking about meal plans

Platform	Access Method	What to Monitor	Signal Examples
LinkedIn	Web scraping (very careful)	Posts, comments, industry discussions	Professional posts about healthy living and productivity
Google Search	SerpAPI or Google Custom Search API	Search result positioning and trends	Search volume for "meal prep Lagos" trending upward

Signal Source 2: Search Intent Analysis

Using the Google Trends API and SerpAPI (or Google Custom Search JSON API), the system monitors what people are actively searching for. It compares trending search queries against the search terms stored in each Product Intelligence Profile.

How it works: Every 6 hours, the system pulls trending and high-volume search queries for each registered product's category and location. The LLM compares these queries against the PIP to determine if there's an actionable match. If a search term is trending that directly relates to a registered product, the Distribution Engine prioritizes SEO content creation for that term.

Signal Source 3: Web Forum & Community Monitoring

Beyond major platforms, the system monitors niche forums, community boards, WhatsApp public channels (via web interface), Telegram public groups, and local classifieds sites like Jiji.ng. These smaller communities often have the highest-intent audiences because people are asking specific, purchase-ready questions.

4.2 The Matching Algorithm

When an intent signal is detected, it must be matched to the right product. The system uses a two-stage matching process.

Stage 1: Semantic Vector Search

The intent signal (the text of the post, comment, or search query) is converted into a vector embedding using the same embedding model used for Product Intelligence Profiles. This embedding is compared against all product embeddings in the vector database using cosine similarity. Products with a similarity score above a configurable threshold (default: 0.75) are considered potential matches.

Stage 2: LLM Verification and Scoring

The top 5 potential matches from the vector search are sent to the LLM along with the original intent signal. The LLM evaluates each match and returns a relevance score

(0-100) and a recommended action. This two-stage approach prevents false positives — the vector search is fast but approximate, while the LLM verification is slower but precise.

Match Output Structure

```
{
  "intent_signal_id": "sig_2026_04_13_001",
  "signal_source": "nairaland",
  "signal_text": "Please I need a good healthy food delivery in Lekki area...",
  "matched_product": "chopfit-lagos-001",
  "relevance_score": 94,
  "recommended_action": "direct_response",
  "recommended_channel": "nairaland_reply",
  "urgency": "high",
  "generated_response_draft": "You should check out ChopFit Lagos..."
}
```

4.3 Intelligence Engine: Continuous Learning

The matching algorithm improves over time through a feedback loop. Every time the Distribution Engine takes an action (posts a response, publishes content, sends outreach), the result is tracked. Did the target click? Did they visit the website? Did they respond positively? Did the post get flagged or removed?

This data is fed back into the system to adjust matching thresholds, prefer certain channels over others, refine the types of signals that lead to conversions, and avoid actions that get flagged or produce negative responses.

5. Layer 3: Autonomous Distribution Engine

This is where the system takes action. The Distribution Engine receives matched intent signals from the Intelligence Engine and executes the optimal response across multiple channels. Each distribution method operates as an independent module that can be enabled, disabled, or tuned per product.

5.1 Distribution Module 1: Conversational Response Engine

What It Does

When someone posts a question or request on social media or a forum that matches a registered product, this module generates and posts a helpful, natural response that includes a recommendation of the product. The response must feel like a genuine recommendation from a real person, not an advertisement.

How to Build It

1. The Intent Signal arrives from the Intelligence Engine with the matched product and the original post.
2. The LLM receives a prompt containing: the original post text, the Product Intelligence Profile, the platform (X, Reddit, Nairaland, etc.), platform-specific tone guidelines (Reddit is casual, LinkedIn is professional, Nairaland uses Nigerian English), and strict instructions to be helpful first and promotional second.
3. The LLM generates a response that answers the person's question or addresses their need, then naturally introduces the product as a recommendation.
4. The response is reviewed by a quality filter (another LLM call with a critic prompt) that checks: does this read as spam? Is the product mention forced? Would this get flagged by moderators? Is it genuinely helpful?
5. If the quality filter passes (score above 80/100), the response is posted through the appropriate platform API or browser automation.

Example Generated Response (Nairaland)

Original post: "Abeg who knows where I fit get good healthy food delivery for Lekki? I don tire to cook after work."

Generated response: "I get you bro! That after-work tiredness no be joke. I've been using ChopFit Lagos for about a month now — they do calorie-counted meals and deliver to

Lekki. The portions are decent and you can pick your meals for the whole week on Sunday. Check their page: www.chopfit.ng. The jollof rice bowl is actually fire."

Critical Rules for This Module

- Never post more than 2-3 responses per platform per day per product. Over-posting triggers spam detection.
- Never use identical or near-identical text across multiple posts. Every response must be uniquely generated.
- Always answer the person's question first. The product mention should feel incidental, not central.
- Include specific details about the product (a specific dish, a specific feature) to make it feel like a real user experience.
- Vary the account profiles used for posting (see Section 7 on Compliance).
- If a platform requires account age or karma to post, maintain aged accounts that participate organically in the community.

5.2 Distribution Module 2: SEO Content Engine

What It Does

For every registered product, the system automatically generates search-engine-optimized articles, blog posts, and web pages targeting the keywords and phrases potential customers are searching for. These pages are published on web properties controlled by the system, so when someone searches for a relevant term, they find content that recommends the product.

How to Build It

1. From each Product Intelligence Profile, extract the top 10-20 search terms.
2. Use SerpAPI or a similar tool to check the current search volume and competition for each term.
3. Prioritize terms with moderate-to-high search volume and low-to-moderate competition.
4. For each prioritized term, send a content generation prompt to the LLM that specifies: the target keyword, the product to recommend, the ideal article length (1,000-2,000 words), SEO best practices (keyword in title, in first paragraph, in headings, natural keyword density), a helpful and informative tone that provides real value beyond just promoting the product.

5. The generated article is reviewed by a quality filter that checks: is this genuinely useful content that someone would want to read? Does the product mention feel natural? Is the SEO structure sound?
6. The article is published to a content site controlled by the system (a blog network hosted on Vercel/Netlify, or a subdomain system).
7. The system submits the new page URL to Google Search Console for indexing via the Indexing API.
8. Monthly, the system checks rankings for each published article and triggers content updates for pages that are ranking on page 2-3 (close to breaking through to page 1).

Content Site Architecture

The system maintains a network of topical content sites. Each site covers a specific niche (health and fitness, technology, business services, etc.) and hosts articles for multiple products within that niche. This is more effective than product-specific microsites because topical authority helps all articles on the site rank better.

- Domain naming: use brandable, niche-relevant domains. For health: something like "FitLifeNaija.com." For tech: "TechEdgeAfrica.com."
- Each site should have 30-50 high-quality articles before it starts ranking well. The system can generate this initial batch within 2-3 days.
- Internal linking between articles on the same site boosts SEO authority. The system should automatically add relevant internal links when new articles are published.
- Each site needs proper technical SEO: sitemap.xml, robots.txt, fast loading speed, mobile-responsive design, and structured data markup (Schema.org).

5.3 Distribution Module 3: Generative Engine Optimization (GEO)

What It Does

When people use AI chatbots like ChatGPT, Claude, Gemini, or Perplexity to ask questions or get recommendations, those AI systems pull information from the web. GEO ensures that the content the system generates is structured in a way that AI systems can pick up and incorporate into their responses.

How to Build It

GEO is not a separate technical module — it is a set of content formatting and structuring principles applied to all content the SEO engine generates. The key principles are:

1. Include direct, concise answers to common questions at the top of each article. AI systems preferentially extract clear, direct answers.
2. Use structured data markup (Schema.org) on all pages — Product schema, FAQ schema, Review schema, LocalBusiness schema. AI systems heavily rely on structured data.
3. Include statistics, specific numbers, and cited facts. AI systems prefer content that includes concrete data over vague claims.
4. Ensure the product name and core value proposition appear in the first 200 words of every article. Many AI systems weight the beginning of content more heavily.
5. Create FAQ pages for each product that directly answer the questions potential customers ask. These are the highest-value pages for GEO because AI systems love pulling from FAQ content.
6. Maintain consistent NAP (Name, Address, Phone) information across all web properties. This is especially critical for local businesses.

5.4 Distribution Module 4: Automated Outreach Engine

What It Does

This module identifies potential customers through public data — social media profiles, business directories, and professional networks — and sends personalized outreach messages to introduce the product. This is the module that most directly replicates the human sales function.

How to Build It

1. The Intelligence Engine identifies potential customer profiles based on public data — people who have expressed relevant interests, follow relevant accounts, or match the ideal customer profile.
2. For each potential customer, the system gathers context: their recent posts, their bio, their apparent interests, and needs.
3. The LLM generates a personalized message that references something specific about the person (their recent post, their bio, their location) and introduces the product in a way that feels relevant to their specific situation.
4. The message is sent through the appropriate channel: email (if the person has a public email), X/Twitter DM (if DMs are open), Instagram DM, or LinkedIn message.
5. A strict sending limit is enforced: maximum 20 outreach messages per day per product, maximum 5 per platform per day. These limits prevent the system from being flagged as spam.

Critical Rules for Outreach

- Every message must be genuinely personalized. No templates. The LLM generates each message from scratch based on the recipient's context.
- The message should provide value even if the person has no interest in the product. Include a useful tip, insight, or resource alongside the product mention.
- Include a clear and easy opt-out. If someone responds asking to not be contacted, their profile is permanently blocklisted.
- Never send outreach to the same person twice unless they responded positively to the first message.
- Comply with applicable regulations. In Nigeria, the NDPR (Nigeria Data Protection Regulation) applies to commercial communications. In Europe, GDPR applies. In the US, CAN-SPAM applies. The system must respect these per the recipient's location.

5.5 Distribution Module 5: Auto-Generated Landing Pages

What It Does

For each registered product, the system automatically generates a dedicated landing page optimized for both human visitors and search engines. This landing page serves as the destination for all traffic the system generates — from social responses, SEO articles, outreach links, and AI recommendations.

How to Build It

- Use a static site generator (Next.js, Astro, or Hugo) to create landing pages from templates.
- The LLM generates the landing page copy based on the Product Intelligence Profile: headline, subheadline, feature list, social proof section (if the business provides testimonials), call-to-action, and FAQ section.
- If the business provided a product image, it is used as the hero image. If not, the system uses a relevant stock image or generates one using an image generation API.
- Each landing page includes: Schema.org Product and LocalBusiness structured data, Open Graph and Twitter Card meta tags for social sharing, a click-to-call or click-to-WhatsApp button (if applicable), Google Analytics or Plausible tracking for conversion measurement, A/B testing capability — the system generates 2-3 headline variants and rotates them.
- Landing pages are deployed automatically to a subdomain system: chopfit.yourdomain.com or yourdomain.com/chopfit-lagos.

6. Platform Compliance & Sustainability

This is arguably the most critical section of the entire blueprint. An AI marketing system that gets banned from platforms is a failed product. Long-term sustainability requires the system to operate within platform guidelines while still delivering value. The goal is not to evade detection — it is to operate in a way that platforms find acceptable.

6.1 Core Compliance Philosophy

The system must provide genuine value to every platform it operates on. If the system's content and responses are helpful, relevant, and well-received by users, platforms have no reason to flag or remove them. The moment the system starts producing low-quality, repetitive, or spammy content, it becomes a target.

The fundamental rule: every action the system takes should make the platform better, not worse. If a response on Nairaland genuinely helps someone find what they're looking for, that's a positive contribution to the platform. If a blog post on the content network provides useful information that answers real questions, Google wants to rank that content. The system only gets into trouble when it prioritizes product promotion over user value.

6.2 Platform-Specific Compliance Rules

X/Twitter

- Use the official X API v2 for all posting and monitoring. Scraping the X website violates their Terms of Service.
- Each account should post no more than 3 product-related responses per day. The rest of the account's activity should be genuine engagement — liking relevant posts, retweeting useful content, posting non-promotional thoughts.
- Maintain an organic posting ratio: for every 1 promotional post, make at least 4-5 non-promotional contributions. This prevents the account from being classified as a bot.
- Never use multiple accounts to amplify the same product simultaneously. This is coordinated inauthentic behavior and is the fastest route to a permanent ban.
- Use unique content for every post. X's spam detection catches duplicate or near-duplicate text across accounts.

Reddit

- Reddit has the strictest anti-spam community. Every subreddit has its own rules about self-promotion.
- Accounts must be at least 30 days old and have positive karma before posting recommendations. Build karma organically through genuine participation.
- Follow the 10% rule: no more than 10% of an account's total posts should be promotional. The other 90% must be genuine contributions.
- Never post the same product link in multiple subreddits within a short time window. Space out posts across days.
- If a subreddit has a specific 'no self-promotion' rule, respect it absolutely. Getting banned from a high-value subreddit is a permanent loss.

Nairaland

- Nairaland is more lenient than Reddit but still flags obvious spam. Write in natural Nigerian English — the LLM should be specifically prompted to use Nigerian colloquialisms.
- Post in the correct board/section. Putting a health food recommendation in the Jobs section will get it flagged.
- Space out posts to no more than 1-2 per board per day.
- Engage with other posts on the platform to maintain the appearance (and reality) of being a genuine community member.

Google (for SEO Content)

- Google's Helpful Content Update specifically targets AI-generated content that exists only to rank in search results without providing real value. Every article must pass the 'would a human find this genuinely useful?' test.
- Avoid thin content — every article should be at least 1,000 words with original insights, not just rephrased product descriptions.
- Include original data, unique perspectives, or specific local knowledge that cannot be found in competing articles.
- Build backlinks naturally by creating content good enough that other sites link to it. Never buy backlinks or participate in link schemes.
- Use Google Search Console to monitor for manual actions or algorithmic penalties. If a content site is flagged, immediately audit and improve the content.

Email (for Outreach)

- Comply with CAN-SPAM, GDPR, and Nigeria's NDPR. Always include an unsubscribe/opt-out mechanism.

- Use proper email authentication: SPF, DKIM, and DMARC records on all sending domains.
- Warm up new sending domains gradually. Start with 10-20 emails per day and increase by 10-20% per week over 4-6 weeks.
- Monitor delivery rates, open rates, and spam complaint rates. If spam complaints exceed 0.1%, immediately pause and investigate.
- Use a dedicated email service provider (SendGrid, Postmark, or Mailgun) that provides delivery monitoring and reputation management.

6.3 Account Management Strategy

The system should not rely on a single account per platform. However, this is not about creating fake accounts — it is about managing multiple genuine brand personas that each serve a specific niche or purpose.

Legitimate Account Structures

- One brand account per content niche. For example, one account focused on health and wellness content, another on technology, another on business services. Each account genuinely engages with its niche community.
- Each account should have a complete profile: bio, profile photo, pinned content, consistent posting history.
- Use different IP addresses for different accounts (residential proxies). Platforms flag multiple accounts operating from the same IP.
- Never share account access patterns. Each account should have slightly different active hours, posting frequency, and engagement patterns.
- Keep a clear record of all accounts and their purposes. This helps manage the system and demonstrates legitimacy if questioned.

6.4 Anti-Detection Technical Measures

These measures ensure the system's automated activities are not flagged by platform bot-detection algorithms. The purpose is not deception — it is to ensure that the system's legitimate, value-providing activities are not incorrectly flagged by automated systems that cannot distinguish between a helpful AI agent and a spam bot.

Browser Fingerprinting

- Use Playwright with the stealth plugin (playwright-extra + puppeteer-extra-plugin-stealth) to prevent headless browser detection.

- Rotate user agents, viewport sizes, and browser language settings to match natural browsing patterns.
- Add human-like delays between actions: random wait times of 1-5 seconds between page loads, 0.5-2 seconds between keystrokes, variable scroll speeds.

Request Patterns

- Never make requests at perfectly regular intervals. Add random jitter of 10-30% to all scheduled tasks.
- Respect robots.txt for all sites. If a site blocks scraping, do not scrape it.
- Implement exponential backoff for rate-limited requests. If a platform returns a 429 (rate limit) error, wait progressively longer before retrying.
- Use residential proxy networks (BrightData, Smartproxy, or IPRoyal) for scraping operations. Datacenter IPs are the first thing platforms block.

Content Variation

- Never generate two pieces of content with more than 30% textual similarity. Use the LLM's temperature parameter (0.7-0.9) and explicit variation prompts to ensure uniqueness.
- Vary sentence structure, opening hooks, product mention positioning, and call-to-action phrasing across all generated content.
- Maintain a content fingerprint database that stores hashes of all generated content. Before publishing, check the new content against existing hashes to prevent accidental duplication.

7. Database Architecture

7.1 PostgreSQL Schema (Core Tables)

businesses

Column	Type	Description
id	UUID (primary key)	Unique business identifier
name	VARCHAR(255)	Business name
email	VARCHAR(255)	Business owner email
plan	ENUM(free, starter, growth, pro)	Subscription tier
created_at	TIMESTAMP	Registration date
status	ENUM(active, suspended, cancelled)	Account status

products

Column	Type	Description
id	UUID (primary key)	Unique product identifier
business_id	UUID (foreign key)	Owning business
name	VARCHAR(255)	Product/service name
description	TEXT	Raw product description
target_customer	TEXT	Target customer description
location	VARCHAR(255)	Service area/location
category	VARCHAR(100)	Product category
website_url	VARCHAR(500)	Business website
image_url	VARCHAR(500)	Product image storage URL
pip_data	JSONB	Full Product Intelligence Profile
status	ENUM(processing, active, paused)	Product marketing status
created_at	TIMESTAMP	Creation date

intent_signals

Column	Type	Description
id	UUID (primary key)	Unique signal identifier
source_platform	VARCHAR(50)	Platform where signal was detected
source_url	VARCHAR(1000)	URL of original post/query
signal_text	TEXT	Full text of the detected signal
signal_type	ENUM(question, need, complaint, search)	Type of intent expressed
detected_at	TIMESTAMP	When the signal was found
processed	BOOLEAN	Whether the matching engine has processed this signal

matches

Column	Type	Description
id	UUID (primary key)	Unique match identifier
signal_id	UUID (foreign key)	The intent signal
product_id	UUID (foreign key)	The matched product
relevance_score	INTEGER (0-100)	AI-assigned match quality
recommended_action	VARCHAR(50)	Action type (respond, content, outreach)
recommended_channel	VARCHAR(50)	Where to take action
status	ENUM(pending, executed, skipped, failed)	Execution status
created_at	TIMESTAMP	Match timestamp

actions

Column	Type	Description
id	UUID (primary key)	Unique action identifier
match_id	UUID (foreign key)	The match that triggered this action
action_type	VARCHAR(50)	Type of action taken
platform	VARCHAR(50)	Where the action was executed
content_text	TEXT	The generated content/response
content_url	VARCHAR(1000)	URL of published content (if applicable)
result_clicks	INTEGER	Clicks generated
result_impressions	INTEGER	Views/impressions

Column	Type	Description
result_conversions	INTEGER	Conversions tracked
flagged	BOOLEAN	Whether the platform flagged the content
executed_at	TIMESTAMP	Execution timestamp

7.2 Redis Usage

- Rate limiting counters: track actions per platform per product per day to enforce compliance limits.
- Job queue: manage the pipeline of signals to process, content to generate, and actions to execute.
- Caching: store frequently accessed PIPs and platform session tokens to reduce database and API calls.
- Deduplication: store hashes of recently processed signals to prevent the system from responding to the same post twice.

7.3 Vector Database Usage

- Store one embedding per Product Intelligence Profile. Update when the PIP is refreshed.
- Store embeddings for recently processed intent signals to enable fast similarity comparisons.
- Use metadata filtering to narrow searches by category, location, and product status before running similarity comparisons.

8. External APIs & Integrations

8.1 Required APIs

API	Purpose	Cost Estimate	Alternative
Claude API (Anthropic)	Content generation, intent analysis, matching, quality filtering	$0.003-0.015 per 1K tokens	OpenAI GPT-4 API, Google Gemini API
Claude API (Anthropic)	Vector embeddings for semantic matching	$0.003-0.015 per 1K tokens	Cohere Embed, sentence-transformers (free/self-hosted)
X/Twitter API v2	Social monitoring and posting on X	Free tier: 1,500 posts/month. Basic: $100/month	Web scraping (higher risk)
Reddit API (via PRAW)	Monitoring and posting on Reddit	Free for personal use	Web scraping (not recommended)
SerpAPI (or suggest free option)	Google search results, trends, keyword data	$50/month for 5,000 searches	Google Custom Search JSON API ($5 per 1,000 queries)
Google Search Console API	Submitting pages for indexing, monitoring rankings	Free	Bing Webmaster Tools API (free)
SendGrid / Postmark	Email outreach delivery	SendGrid free tier: 100/day. Paid: $15/month	Mailgun, Amazon SES
BrightData / Smartproxy	Residential proxies for web scraping	$5-15/GB of traffic	IPRoyal, Oxylabs
Vercel / Netlify	Hosting auto-generated content sites and landing pages	Free tier available. Pro: $20/month	AWS S3 + CloudFront, Cloudflare Pages
Pinecone	Vector database for semantic matching	Free tier: 100K vectors. Paid: $70/month	Weaviate (self-hosted, free), ChromaDB (open source)

8.2 Cost Projection: Running 100 Products

At 100 registered products with moderate activity:

Service	Monthly Cost Estimate
LLM API (Claude/OpenAI) — ~500K tokens/day for generation and analysis	$45-75
Embeddings API — ~50K tokens/day for matching	$1-3
X API Basic tier	$100
SerpAPI — 5,000 searches/month	$50
Residential Proxies — ~10GB traffic/month	$50-100
Hosting (Vercel Pro)	$20
Email Service (SendGrid Pro)	$15
Vector Database (Pinecone Starter)	$0-70
Server Infrastructure (VPS or cloud)	$40-100
Total Estimated Monthly Cost	$321-533

At this cost, each product costs approximately $3-5/month in infrastructure. If you charge even $10/month per product on the free tier and $30-50/month on paid tiers, the unit economics work from day one.

9. MVP Build Plan: 90-Day Roadmap

9.1 Phase 1: Foundation (Days 1-30)

Week 1-2: Core Infrastructure

- Set up the backend server (FastAPI or Express) with PostgreSQL and Redis.
- Build the business registration and product submission API endpoints.
- Build the Product Intelligence Profile generation pipeline (LLM prompt + JSON parsing + database storage).
- Set up the vector database and embedding generation pipeline.
- Build a basic web dashboard where businesses can register and submit products.

Week 3-4: Intelligence Engine v1

- Build the social listening module for one platform (start with X/Twitter using the official API).
- Build the search intent analysis module using SerpAPI.
- Build the two-stage matching algorithm (vector search + LLM verification).
- Create the match output queue that feeds into the Distribution Engine.
- Set up the feedback loop database tables and basic tracking.

9.2 Phase 2: Distribution (Days 31-60)

Week 5-6: First Distribution Modules

- Build the Conversational Response Engine for X/Twitter (auto-reply to matched signals).
- Build the SEO Content Engine (article generation + publishing to one content site).
- Build the auto-generated landing page system.
- Implement quality filters for all generated content.

Week 7-8: Expand Monitoring

- Add Nairaland monitoring (web scraping with Playwright).
- Add Reddit monitoring (PRAW integration).
- Add the Automated Outreach Engine (email only, starting with 10 messages/day/product).
- Build the compliance rate-limiting system across all platforms.

9.3 Phase 3: Polish & Launch (Days 61-90)

Week 9-10: Dashboard & Analytics

- Build the business-facing analytics dashboard showing: active channels, signals detected, actions taken, clicks, impressions, and leads generated.
- Add real-time status updates so businesses can see their product's marketing activity live.
- Build the billing system with free tier (1 product, basic distribution) and paid tiers.

Week 11-12: Testing & Launch

- Run 10-20 beta products through the system and gather data on match quality, response rates, and platform compliance.
- Tune the matching threshold and quality filters based on beta results.
- Fix compliance issues discovered during beta testing.
- Launch publicly with a clear value proposition: 'Enter your product. AI does your marketing. No ad budget needed.'

10. Monetization Strategy

10.1 Pricing Tiers

Tier	Monthly Price	Products	Features
Free	$0 (or ₦0)	1 product	Basic SEO content generation, 1 content site, landing page, basic analytics
Starter	$15/month (₦25,000)	3 products	All Free features + social monitoring (2 platforms), conversational responses, weekly performance report
Growth	$40/month (₦65,000)	10 products	All Starter features + all platform monitoring, email outreach, GEO optimization, daily performance updates, priority matching
Pro	$100/month (₦165,000)	Unlimited products	All Growth features + API access, custom content sites, white-label landing pages, dedicated support, advanced analytics

10.2 Additional Revenue Streams

- Performance bonus model: an optional add-on where businesses pay a small fee per qualified lead generated (pay-per-lead rather than pay-per-click). This aligns incentives — the system only earns more when it delivers actual results.
- Content marketplace: allow businesses to purchase additional SEO articles or landing page variations beyond what their tier includes.
- API access: let developers and agencies integrate the matching engine into their own tools for a monthly API fee.
- Enterprise partnerships: custom white-label deployments for marketing agencies who want to offer AI-powered marketing to their own clients.

11. Key Technical Decisions & Recommendations

11.1 Start with Claude API

For the LLM layer, Claude (Anthropic) is recommended as the primary model for several reasons: it handles nuanced instructions well (critical for generating platform-appropriate content), it has strong JSON output formatting (essential for structured PIP generation), and it maintains quality at high volume. Use claude-sonnet-4-20250514 for most tasks and claude-haiku-4-5-20251001 for high-volume, lower-complexity tasks like quality scoring.

11.2 Build the Monitoring Layer First

Before building any distribution modules, perfect the Intelligence Engine. If the system cannot accurately detect intent signals and match them to the right products, every action it takes will be wasted or — worse — harmful to the product's reputation. Spend 60% of development time in Phase 1 on the monitoring and matching accuracy.

11.3 Instrument Everything

Log every action, every match, every response, and every result. This data is the system's most valuable asset. It feeds the learning loop, proves ROI to businesses, enables debugging when things go wrong, and provides the data needed to continuously improve the matching algorithm.

11.4 Legal Considerations

- Register a company for the product. Operating commercial software that interacts with third-party platforms and sends commercial messages requires a legal entity.
- Create clear Terms of Service that businesses agree to when they register. These should state that the system operates within platform guidelines and that businesses are responsible for the accuracy of their product information.
- Implement a content review process (even if automated) that catches misleading product claims before they're published. You do not want the system marketing a fraudulent product.
- Comply with Nigeria's NDPR, Europe's GDPR, and US CAN-SPAM for any data collection or email outreach. Build these compliance checks into the system from day one.

11.5 Scaling Path

The system described in this document can serve 100-500 products on a single server. Beyond that, the architecture needs to scale horizontally. Key scaling points: separate the monitoring workers from the distribution workers so they can scale independently, use a message queue (RabbitMQ or Amazon SQS) between layers so the system can handle traffic bursts, implement database read replicas for the analytics dashboard, and cache frequently accessed PIPs in Redis to reduce database load.

End of Technical Blueprint

Built for Aniekan Israel — Light to the World. Massive Thoughts. Massive Execution.